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Modeling the Age-Structured Dynamics of Pandemic Spread: A Comprehensive Approach

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Introduction

Originating in China, the COVID-19 pandemic has fundamentally altered our economy, health systems, and social behaviors across the globe. While outbreaks of swine flu (H1N1) or malaria have occurred, no modern disease has spread as widely as COVID-19 has. With a global death toll of nearly 7 million individuals and nearly 800 million confirmed cases of COVID-19, the disease continues to infect individuals across the globe.

Indeed, recent studies have suggested that factors like globalization and climate change are contributing to the possibility that 'extreme pandemics' like COVID-19 could emerge more frequently in the future. Such pandemics would have the potential to infect hundreds of millions of individuals. COVID-19 itself continues to evolve rapidly, presenting new strains and variants every few months that require booster shots and continued social distancing measures to safeguard against.

As such, it is crucial to employ new models that better reflect novel diseases that may emerge in the future. By examining COVID-19, we can better prepare for such diseases. In particular, we seek to remedy oversimplifications of particular models that render them unsuitable for COVID-19 pandemics. In particular, we seek to explore how age classes impact the game theory model of vaccination and disease transmission. Through the introduction of 3 age classes (Juvenile, Adult, and Elderly), each with different transmission rates and behaviors, we hope to more accurately model diseases like COVID-19.

Lastly, we hope to make a policy recommendation for age-group vaccination efforts, and answer the following questions: Which age group is the most advantageous to target vaccination efforts in order to mitigate the spread of similar pandemics and/or death rates? How should vaccination efforts evolve throughout the course of pandemic stages?

Significance

While the current methods that are used for modeling the spread of diseases work well, there are a multitude of simplifications that make these models far too theoretical to reflect real world disease dynamics. We seek to add specificity to the model by eliminating some of these oversimplifications, such as the idea that the transmission rate between individuals is the constant for every individual. We hope that by rectifying these oversimplifications, we can present a more comprehensive and realistic model of disease dynamics that accounts for nuances. Doing so has many important applications, such as determining which age class is the most important spreader of the disease in order to effectively target vaccination efforts. As mentioned before, this would allow for improved modeling for rapidly aging populations. Given that current SIR models do not take into account age classes, we expect that our model will allow for more refined recommendations in the face of a pandemic.

Rationale

As previously mentioned, current SIR models for COVID-19 are somewhat insufficient in that they fail to account for age-structure, instead opting for average values when analyzing α, γ and β. Previous examples of SIR models we considered were those proposed by Cooper, Mondal, & Antonopolous (2021) as well as Din & Algehyne (2021).[1] The model put forward by Cooper et al. (2021) doesn't employ traditional α, γ and β values within their ODEs, instead opting for a and b values that represent exponential growth and decay values, respectively. We've found the lack of complexity within this model as well as the choice of a and b values in place of more traditional parameters provides an opportunity to expand on the model by substituting in the traditional SIR parameters and adding specificity. Furthermore, the model put

[1] Reference Appendix A and Appendix B for Models

forward by Din & Algehyne (2021), while more specific than Cooper et al., utilizes average α, γ and β values that don't adequately account for the empirical deltas between age classes.

Therefore, we feel the addition of age-structures is a necessary facet of new SIR models because age classes differ in the following significant manners: (1) contact rates, (2) recovery time, (3) mortality rates, (4) infection rates, and (5) vaccination perceptions. Recent studies have demonstrated that individuals' physical contact rates vary dramatically with age (Zhaoyang, Sliwinski, Martire, & Smyth, 2018). For instance, older adults tend to have fewer social contacts than younger individuals, which may make them less susceptible to infection but also limit their access to social and economic resources. Recovery time is significantly longer for older individuals infected with COVID-19, which may extend their infectious period and increase the likelihood of transmission (Voinsky, Baristaite, & Gurwitz, 2020). Mortality rates are also significantly higher for elderly and immunocompromised individuals than younger ones (Bonanad et al., 2020). Therefore, it is essential to reflect the heterogeneity in these factors within updated SIR models to develop effective strategies to control the spread of diseases like COVID-19.

Additionally, vaccination issues are informed by costs and payoffs. The rise of misinformation throughout the United States has contributed to 'patches' of anti-vax communities, which has been historically seen in cases like smallpox or measles, with anti-vax sentiments stratified by age demographic. The Department of Health and Human Services reports demographic discrepancies for vaccine hesitancy (Beleche et al., 2021). COVID-19 is also different in that the digital advent has given citizens access to more information. As such, we seek to add a game theory element to our model in order to take into consideration the differences between how age classes behave in an epidemic. Specifically, we reference notions

from replicator equations that can calculate the “expected vaccine coverage level” (Bai, 2019). Thus, it’s crucial to account for differences in age classes when modeling future pandemics that are similar to COVID-19. For notable diseases like Ebola or Monkeypox, mortality or infection rates are not as drastically different between age classes as they are for COVID-19. We plan to include these factors in our new SIR model.

Methodology

In this section, we elucidate the chosen methodology for this study as well as the reasoning behind our model and parameters. To briefly echo notions mentioned earlier in this work, we expand on existing SIR models by adding age-structure and game theory components in order to present a more comprehensive analysis of both disease spread as well as vaccination rates throughout a population.

Contact Matrix

Existing models assume a constant interaction rate between individuals in a population. However, the addition of age-structures necessitates a more nuanced approach to interaction rates as intuitively, juveniles are more likely to interact with juveniles than elderly individuals because of schools or sports teams, for example. To model the interactions between age classes, we developed a matrix that describes the rate of contact between three different age classes: juveniles (0 to 20 years old), adults (21 to 60 years old), and elderly people (61 to 80+ years old). The data was taken from papers that project social contact matrices through surveys and demographic data (Prem and Cook (2017)). Each entry in the matrix represents the relative number of interactions that an individual of each corresponding age group has in a day.

The original contact rate matrix came from Ram and Schaposnik (2021). This gave us relative contact rates for age classes from age 0 to 80+ in 10 year increments. We took the average contact rates over our extended age classes to obtain a beta matrix of the three age classes.

This resulted in a matrix with three columns and three rows, with the columns being juvenile, adult, elderly (left to right) and the rows being juvenile, adult, elderly (top to bottom).

But, this matrix assumes that the age distribution is constant across the three age classes. To further increase accuracy, we wanted to account for the fact that the amount of people in each of our constructed age classes is not the same. To do this, we assumed a uniform distribution of individuals within each of our age classes, but not across age classes.

Using data from the U.S. Census Bureau, we were able to find the approximate proportion of the population in each of our age classes. We found that juveniles make up about 33.4% of the US population, adults make up 52.9% of the US population, and elderly people make up 13.7% of the US population (U.S. Census Bureau (2022)). Then, using the contact rate matrix that we produced previously, we multiplied each element by the probability of that interaction occurring (See Figure 1). For instance, while calculating the probability that a juvenile would interact with an adult, we would take the corresponding value from the averaged matrix produced by Ram and Schaposnik (2021) and multiply it by the product of 0.334 and 0.529.

$$\beta = \begin{array}{c} \begin{array}{ccc} \text{Juvenile} & \text{Adult} & \text{Elderly} \end{array} \\ \begin{pmatrix} 1.20 & 0.68 & 0.10 \\ 0.68 & 2.49 & 0.24 \\ 0.10 & 0.24 & 0.30 \end{pmatrix} \end{array} \begin{array}{l} \\ \text{Juvenile} \\ \text{Adult} \\ \text{Elderly} \end{array}$$

Figure 1: The relative contact rates between specific age classes, accounting for the fact that each age class comprises a different proportion of the total US population.

Rewriting Differential Equations

To transition from a conventional SIR model, additional considerations had to be taken. Firstly, we had to consider a constant flow of aging-up between the three age classes, in which a proportion of individuals in one age class moves to the next in a given year.

The strength of the age-based SIR model is that it allows us to parameterize each class differently. We derived separate death rates, perceived costs, and sensitivities for each age class based on a comprehensive literature review with statistics from the COVID-19 pandemic.

We also expanded on traditional SIR models by considering vaccination as a measure of pandemic preparedness.

$$\frac{dS_J}{dt} = (1 - v_J)\mu - \sum_{j=1}^{3} \beta_{Jj} S_J I_j - \alpha_A S_J - \mu S_J$$

$$\frac{dI_J}{dt} = \sum_{j=1}^{3} \beta_{Jj} S_J I_j - \gamma_J I_J - \alpha_A I_J - \mu I_J$$

$$\frac{dR_J}{dt} = v_J \mu + \gamma_J I_J - \alpha_A R_J - \mu R_J$$

$$\frac{dS_A}{dt} = (1 - v_A)\alpha_A S_J - \sum_{j=1}^{3} \beta_{Aj} S_A I_j - \alpha_E S_A - \mu S_A$$

$$\frac{dI_A}{dt} = \alpha_A I_J + \sum_{j=1}^{3} \beta_{Aj} S_A I_j - \gamma_A I_A - \alpha_E I_A - \mu I_A$$

$$\frac{dR_A}{dt} = v_A \alpha_A S_J + \alpha_A R_J + \gamma_A I_A - \alpha_E R_A - \mu R_A$$

$$\frac{dS_E}{dt} = (1 - v_E)\alpha_E S_A - \sum_{j=1}^{3} \beta_{Ej} S_E I_j - \mu S_A$$

$$\frac{dI_E}{dt} = \alpha_E I_J + \sum_{j=1}^{3} \beta_{Ej} S_E I_j - \gamma_E I_E - \mu I_E$$

$$\frac{dR_E}{dt} = v_E \alpha_E S_A + \alpha_E R_A + \gamma_E I_E - \mu R_E$$

$$\frac{dv_k}{dt} = K v_k (1 - v_k)(-c_{v,k} + c_{I,k} \sum_{j=1}^{3} \beta_{kj} I_J), \forall k \in K = \{J, A, E\}$$

α_k: proportion of same-infection-status individuals ageing from age class k - 1 to k
γ_k: recovery rate of age class k
μ: birth / death rate
c_v: perceived cost of vaccination
c_I: perceived cost of infection
β_{ij}: contact rate between age classes i and j
K: sensitivity of vaccination rate to new information

Figure 2: Each age class (J = juveniles, A = adults, E = elderly people) has three associated differential equations to describe susceptible, infected, and recovered frequencies. There is an additional set of differential equations that describes the proportion of vaccinated individuals in each age class.

Parameters and Assumptions

To determine the parameters and constants of our model, we conducted an extensive literature review. For parameters where data was limited, we estimated numbers based on historical pandemics or basic assumptions. These parameters were also varied between scenarios to model different outcomes. The parameters are defined as follows:

1. α_k is the proportion of individuals in a given age class that age up to the next. We calculated this by taking the reciprocal of the number of ages represented within a single age class. So, for Juveniles (age 0 to 20 years), α_k is given by 0.05.
2. γ_k is the recovery rate for each age class. This was estimated and weighted based on data collected from Ram and Schaposnik (2021). The literature review yielded no results as to the age-delineated recovery rates, but we were able to determine that the overall recovery rate is about 97% across all of the people who got COVID-19 (Elflein (2023)). So, we assumed a 98% recovery rate for Juveniles and Adults, and an Elderly recovery rate that is lower than the other two (90%).
3. c_v and c_i represent the perceived costs of vaccination and infection for each age class. They are unitless and only have meaning when respective to each other. We assumed that members of the elderly age class have a higher perceived risk of infection than members of the other age classes, as a function of their rationality and past experience with illness.

4. β_{ij} values are given by the contact matrix we developed.
5. K tunes the sensitivity to new information for the general population. This represents a wide variety of factors, like confidence in media or news dissemination. Under the framework of collective behavior, we can think of varying K as varying the degree to which individuals follow others rather than operating on their own signal.
6. We assumed a death rate of zero from infections for the duration of our simulation.
7. We assumed that the population size remained constant, and normalized the population to a value of 3 (i.e. the sum of susceptibles, infectives, and recovered across all age classes sum to 3).

Results

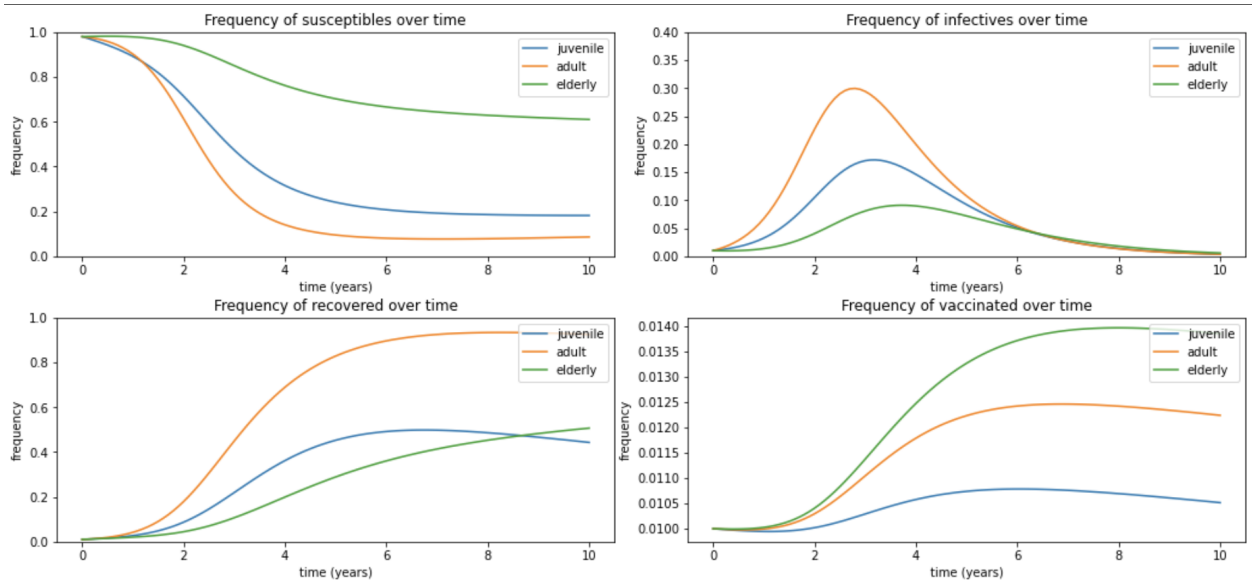

Figure 3: The control scenario, in which we assume a small initial proportion of individuals are vaccinated at the onset of an infectious outbreak.

In Figure 3, the control scenario, we assume that only a small proportion of individuals are vaccinated at the outset. This scenario is reflective of the early stages of the COVID-19 pandemic, when many protective measures were absent or in their nascence - vaccines, safety protocols, mask mandates, etc. It is thus reflective of general unpreparedness to the disease. Due to differential contact rates and recovery rates, we observe differences in the extent to which age classes are infected by the disease and the rate at which they recover from the disease. Because contact rates are highest for the adult age class, and recovery rates are highest for juveniles and adults, the adult age class is infected the fastest and recovers the fastest. Conversely, while elders do not interact with each other and with other age classes as often as other age classes do, they are more prone to prolonged illnesses and recovery times to disease, so the elderly age class is infected the slowest but also recovers the slowest. We also see that, because elders have a much

higher perceived risk of infection than other age classes, the elderly age class has the highest vaccination uptake.

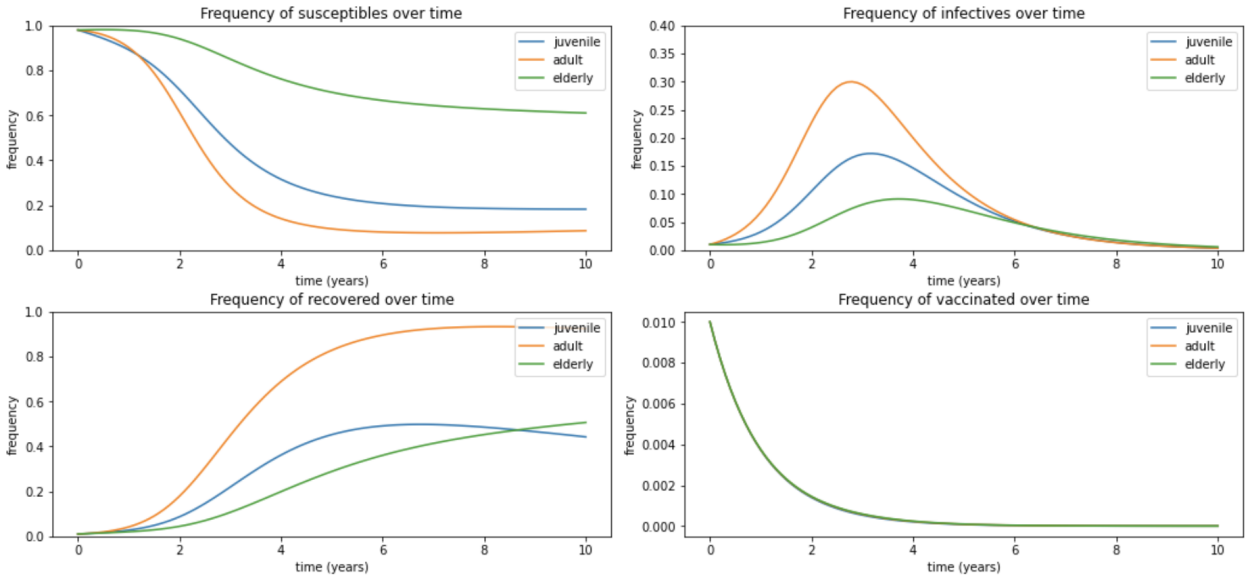

Figure 4: The fear of vaccination scenario, in which all three age classes perceive an increased risk of vaccination.

In Figure 4, the fear of vaccination scenario, we simulate a fear of vaccination, with a notable example being ongoing anti-vaccination sentiments to the COVID-19 vaccine. The dynamics of this scenario look very similar to those of the control scenario. However, we see that there is a constant decline in the frequency of vaccinated individuals over time, in comparison to the rise then fall of the frequency of vaccinated in the control. Though the implications of anti-vaccination sentiments are not immediately clear, as the evolution of the disease does not seem to be discernibly different in this scenario relative to the control, we can think of the feasible real-life scenario in which the disease once again appears at a later time, and because of anti-vaccination sentiments we are again poorly prepared to handle it.

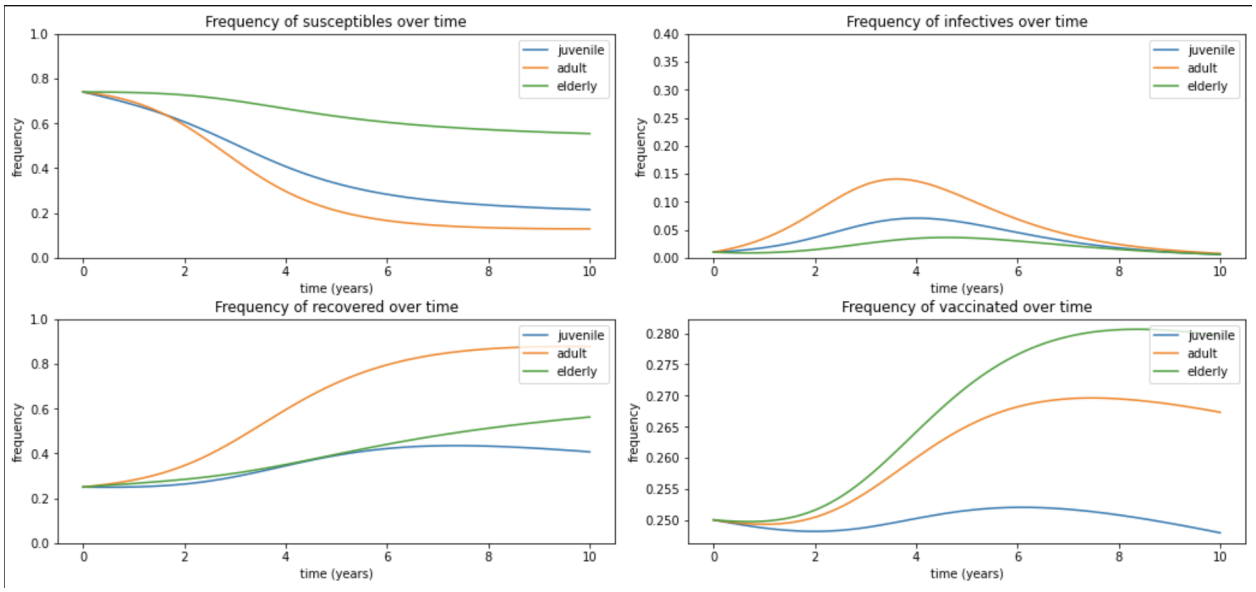

Figure 5: The proactive vaccination scenario, in which a greater proportion of individuals are vaccinated prior to the infectious outbreak.

In Figure 5, the proactive vaccination scenario, we simulate general preparedness and a proactive approach to epidemic control, whereby a large proportion of individuals are vaccinated ahead of time. Here, we see a remarkable improvement in the control of the disease. Herd immunity is reached at a much lower proportion relative to the control scenario. For instance, in this scenario, the adult age class appears to reach herd immunity at approximately half the frequency of the control. This improvement in the dynamics of the disease cannot be overstated, as even though both the control scenario and proactive vaccination scenario appear to lead to a disease-free equilibrium, reducing the peak frequency of infectives at herd immunity averts many of the problems around hospital overflow that we had seen in the COVID-19 pandemic, and reduces the suffering and potential mortality of individuals that would have been infected with the disease if not for proactive vaccination and herd immunity.

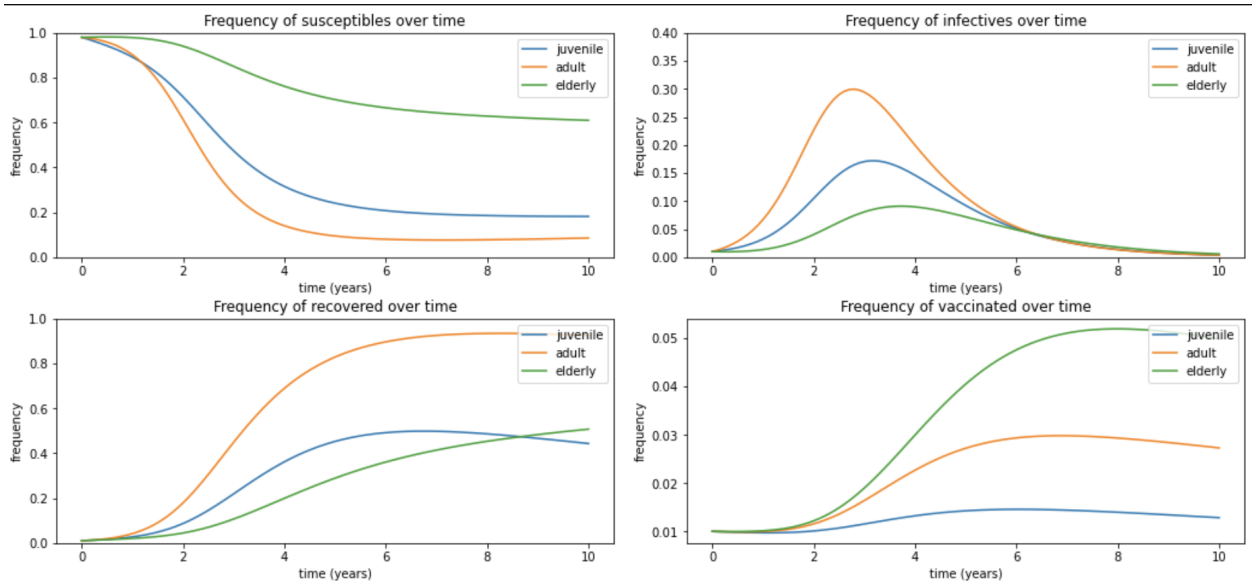

Figure 6: The heightened sensitivity to information scenario, in which individuals are more sensitive to the vaccination behaviors of others via word of mouth, news sources, or otherwise.

In Figure 6, the heightened sensitivity to information scenario, we consider a population in which individuals are more sensitive to the vaccination behaviors of others. Thus, collective behavior and group dynamics play a larger role in the extent to which individuals get vaccinated. As a whole, individuals get vaccinated more quickly in response to the epidemic, leading to a higher frequency of vaccinated individuals over time relative to the control.

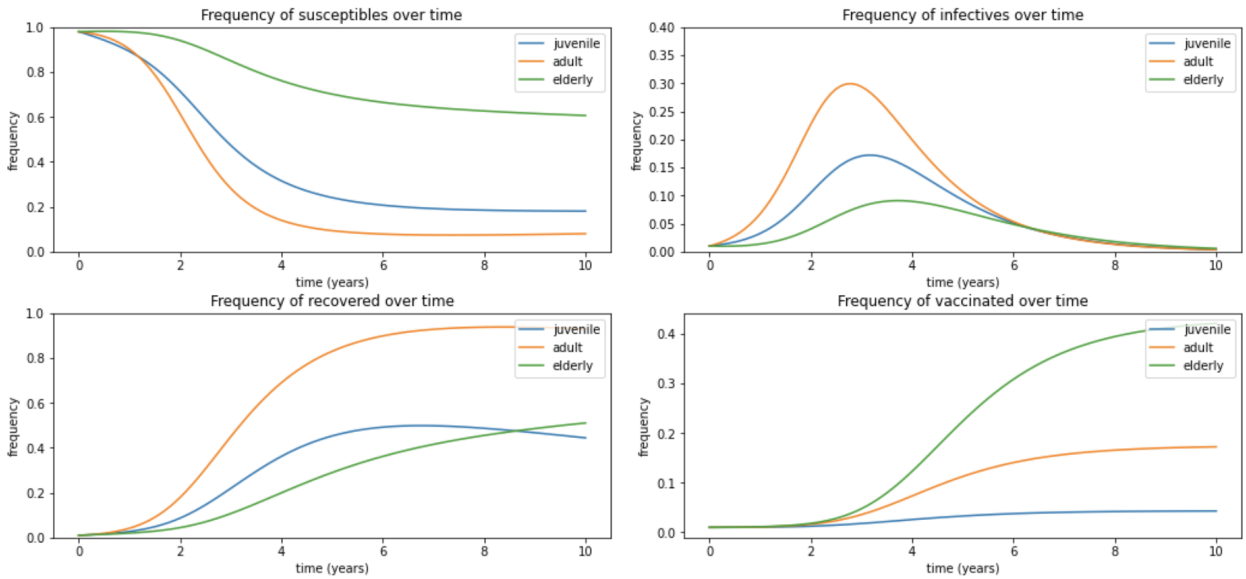

Figure 7: The better education scenario, in which individuals are better educated about the disease and vaccines.

In Figure 7, the better education scenario, individuals are more informed about the risks of infection and the risks of vaccination. Perceived costs approach true costs, leading to a decrease in the perceived cost of vaccination and an increase in the perceived cost of infection. As a result, individuals get vaccinated at a much faster rate in all age classes.

Discussion

Implications

In the graphs above, we simulate the dynamics of the disease and vaccination over a span of ten years, under a variety of scenarios. Note that we do not specify the limits of the y-axis for the frequency of vaccination as the graphs become visually uninterpretable at low levels of vaccination.

Given the remarkable inelasticity of disease evolution in all of the scenarios other than the proactive vaccination scenario, our model and results suggest that perhaps one of the best ways to deal with an epidemic is to be proactive rather than reactive with our vaccination efforts.

That being said, we see how much vaccination rates can vary from scenario to scenario based on behavioral elements like risk perceptions around vaccines and infection. For example, we see that better education about the disease and vaccine can greatly increase vaccination rates, which although may not yield much benefit during the current outbreak, could certainly prepare the population for future outbreaks. In a sense, reactive vaccination efforts in the current outbreak can be reframed as proactive vaccination efforts for future outbreaks.

Crucially, we introduce two components to our model over and above the traditional SIR model - age structure and varying risk perceptions. First, by including age structure, we can better model differences in behavior, recovery rates, and so forth between age classes. For instance, the Journal of the American Medical Association found that COVID-19 recovery time is longer for elderly patients relative to other patients (Wirth et al. (2020)), which we are able to incorporate into our model by virtue of our age structure extension. Second, by including varying risk perceptions, we can model vaccination rate as a dynamic quantity that changes over time, as individuals evaluate and reevaluate the competing costs of vaccination and infection.

These competing interests connect more broadly to the game theoretic nature of vaccination. Each individual can choose between vaccinating and not vaccinating, and their choice depends on risk perceptions around the disease and the vaccine. Though our model takes the perceived cost of vaccination to be fixed, in reality, the perceived cost of infection increases as the total frequency of infections in the population increases. So, at sufficiently low levels of infection, individuals are more inclined not to vaccinate, as they perceive the risk of contracting the disease to be sufficiently low relative to the risk of vaccination. Further, an individual can "free-ride" on the vaccination efforts of others via herd immunity and endure the entire epidemic without ever vaccinating, provided that sufficiently many others vaccinate. Taking into account that these individualistic incentives are important to studying how vaccination rates can change over time, researchers and policymakers can influence the trajectory of vaccination uptake with tools like vaccine mandates and better education.

Limitations

While our model is more finely tuned than general SIR models, there are still limitations that are inherent to our model.

Firstly, our model is delineated by three age classes. This gives only three opportunities for vaccination, and severely limits the characteristics of each class. A parent that chooses to vaccinate their teenager may not make the same choice for their infant. In reality, factors like mortality, vaccine hesitancy, and contact rates are a spectrum that varies finely across all ages.

Secondly, our model may not account for specific characteristics of the COVID-19 pandemic. We fail to consider issues like multiple (competing) strains, multiple vaccinations, and multiple infections. Including these would severely complicate our model.

Lastly, our model inherits issues that are also common to general SIR models. These may include factors like population density, patches of vaccine hesitancy, or varying recovery times.

Conclusion

Different age groups are clearly affected differently in the presence of a pandemic, due to both variable infection rates, recovery rates, and risk perception. We analyzed COVID-19 and its effects on three subgroups of the US population: juveniles (age 0 to 20), adults (age 21-60), and elderly people (age 61-80+). In all of the scenarios we investigated, adults tended to get infected the fastest, as they have high contact rates, while elderly people, who are the most at-risk of all the age groups, tend to get infected at a slower rate than the other age classes but recover more slowly as well. It was also found that proactive vaccination efforts hugely reduce the amount of infected individuals in all three ages classes over time.

Proactive vaccination appears to be the best way to prevent large disease outbreaks. While we also modeled situations in which there was heightened sensitivity to the actions of others and where there was an increase in education about the costs of vaccination and infection, we see that the number of infected individuals is very similar to the control scenario. This indicates that these changes in sensitivity and education produce negligible results if the disease has already progressed to a certain extent.

Therefore, to minimize the spread of COVID-19, we recommend that the age class that should be prioritized in vaccination efforts is adults. This will slow the spread of disease, as they have the highest contact rates, and their infection rate increases the quickest. This will give more time to the entire population to react accordingly. However, it might be argued that the elderly population should be vaccinated first as they have the highest mortality rate. But, this then allows the disease to spread amongst the rest of the population at a very rapid rate.

In order to minimize deaths due to COVID-19, it would be best to vaccinate the elderly people first. This will result in a larger amount of people getting COVID-19 overall, but will

decrease the amount of elderly people who are infected and thus reduce the amount of deaths due to disease.

In the future, we hope to employ this model to better inform health policy for other extreme pandemics that may emerge. This model provides a much more fine-grained analysis of pandemics. However, while this model does attempt to present a more accurate version of the already-existing SIR models, there are still more factors that it does not necessarily account for that can further improve its accuracy. For instance, as we have seen with COVID-19, there can be multiple strains infecting people at once, each with different transmission rates, recovery rates, and mortality rates. Additionally, diseases tend to exhibit seasonality in their transmission rates, which is not accounted for in this model; however, we hope that by adding specificity to existing SIR models, we can increase accuracy and responsiveness of vaccination efforts to prevent unnecessary infections and deaths from occurring during times of pandemic.

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Appendix A.

Cooper, Mondal, & Antonopolous (2020) SIR Model

This appendix constitutes the SIR employed for the aforementioned study. It describes the time evolution of the following three populations:

1. Susceptible individuals, *S(t)*: These are those individuals who are not infected, however, could become infected. A susceptible individual may become infected or remain susceptible. As the virus spreads from its source or new sources occur, more individuals will become infected, thus the susceptible population will increase for a period of time (surge period).
2. Infected individuals, *I(t)*: These are those individuals who have already been infected by the virus and can transmit it to those individuals who are susceptible. An infected individual may remain infected, and can be removed from the infected population to recover or die.
3. Removed individuals, *Rm(t)*: These are those individuals who have recovered from the virus and are assumed to be immune, *Rm(t)* or have died, *D(t)*.

$$\begin{aligned} \frac{dS(t)}{dt} &= -aS(t)I(t), \\ \frac{dI(t)}{dt} &= aS(t)I(t) - bI(t), \\ \frac{dR_m(t)}{dt} &= bI(t), \end{aligned}$$

Appendix B.

Din & Algehyne (2021) SIR Model

This appendix constitues the SIR model employed by the aforementioned authors in their mathematical analysis of the spread of COVID-19.

$$\frac{dS(t)}{dt} = b - k(1 - \alpha S(t)I(t)) - \alpha k\beta S(t)I(t) - \mu S(t)$$
$$\frac{dI(t)}{dt} = k(1 - \alpha S(t)I(t)) + \alpha k\beta S(t)I(t) - (d_0 + \gamma + \mu)I(t)$$
$$\frac{dR(t)}{dt} = \gamma I(t) - \mu R(t).$$

The model employs the following parameters whose meanings are elucidated below for comprehensive understanding:

$S(t)$ Susceptible compartment

$I(t)$ Infected compartment

$R(t)$ Recovered compartment

$d0$ Death due to corona

μ Natural death

b Birth rate

β Protection rate

k Constant rate

α Isolation rate

γ Recovery rate

Note the evident lack of age-structure or game theory present in the model, which we leverage for added specificity in this study.